Approach: leakage-safe direct forecasts

We frame the challenge as tabular time-series regression, compare the required model families, and select models with rolling-origin CV.

1 Direct forecast targets

For horizon h, target = BikeCount(t+h). We evaluate one-hour and next-day hourly forecasts.

2 Required model families

Ridge covers linear regression, RandomForest / XGBoost cover tree-based models, and MLP covers neural networks.

3 Time-series validation

Rolling-origin CV uses chronological 14-day forecast windows for model selection.

Feature engineering

- **Calendar** target hour, weekday, month, weekend
- **Cyclical** sin/cos encodes hour and month cycles
- **Weather** target-hour weather buckets + measurements
- **BikeCount** horizon-aware lags and rolling means only

Leakage controls & assumptions

- **BikeCount** horizon-aware lags from known history
- **Weather** target-hour values treated as exogenous input
- **Validation** chronological rolling-origin CV

8,759 clean rows

7 model candidates

14 CV windows / horizon

From model comparison to final XGBoost

We first compared simple baselines and required model families, then selected the strongest cross-validation performer.

Selection funnel

1 Compare model families

Ridge, RandomForest, XGBoost and MLP cover the required linear, tree-based and neural-network approaches.

2 Identify strongest candidate

XGBoost is strongest across the rolling validation windows.

3 Refine XGBoost

Test XGBoost variants, including adjusted tree settings and log1p(BikeCount).

Rolling CV MSE	+1h	+24h
NAIVE BASELINES		
naive horizon lag	25,688	56,681
naive lag168	22,816	22,816
MODEL FAMILY SCREENING		
Ridge	10,803	19,096
RandomForest	5,263	14,681
MLP	7,657	22,316
XGBoost	4,459	14,583
XGBOOST REFINEMENT		
XGBoost log1p	4,652	19,569
XGBoost tuned log1p	4,686	16,534
XGBoost tuned	**4,370**	**13,890**

Takeaway **Boosted trees fit this tabular time-series task best; XGBoost tuned achieves the lowest rolling validation error on both horizons.**